

KURGAN, Russian Federation

WMO: 286610

Lat: 55.4562N

Lon: 65.4012E

Elev: 72

StdP: 100.46

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.6	(c) -29.5	(d) -35.6	(e) 0.1	(f) -32.5	(g) -32.6	(h) 0.2	(i) -29.3	(j) 10.4	(k) -7.6	(l) 9.5	(m) -8.6	(n) 2.2	(o) 180	(p) 0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.9	(c) 31.5	(d) 19.3	(e) 29.6	(f) 18.9	(g) 27.8	(h) 18.3	(i) 21.2	(j) 28.4	(k) 20.3	(l) 27.2	(m) 19.3	(n) 25.6	(o) 4.3	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	13.7	23.7	17.9	13.0	22.9	17.0	12.2	22.2	61.7	28.5	58.7	27.1	55.4	25.5	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.2	(b) 9.0	(c) 8.0	(e) -35.3	(f) 34.0	(g) 3.6	(h) 1.8	(i) -37.9	(j) 35.4	(k) -40.0	(l) 36.4	(m) -42.1	(n) 37.5	(o) -44.7	(p) 38.8		
			DB													
			WB													
				-35.4	22.7	3.6	1.4	-38.0	23.6	-40.1	24.4	-42.1	25.2	-44.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-16.2	-14.3	-5.7	5.1	13.1	17.9	19.3	17.4	11.4	4.0	-5.7	-12.9		
	DBStd	13.96	7.07	7.08	5.51	5.70	5.29	4.58	3.90	4.17	4.70	4.81	7.33	7.54		
	HDD10.0	3592	811	679	487	167	30	3	0	1	38	193	471	711		
	HDD18.3	5817	1069	912	746	398	177	61	36	68	214	444	721	969		
	CDD10.0	992	0	0	0	19	125	241	289	231	79	8	0	0		
	CDD18.3	176	0	0	0	1	13	49	67	41	5	0	0	0		
	CDH23.3	1946	0	0	0	9	199	578	699	393	67	0	0	0		
	CDH26.7	637	0	0	0	1	53	203	234	129	17	0	0	0		
(14) Wind	WSAvg	3.7	3.4	3.6	4.0	4.2	4.4	3.8	3.4	3.4	3.6	3.6	3.7	3.5		
(15) Precipitation	PrecAvg	380	20	12	13	20	33	54	59	55	36	31	26	21		
	PrecMax	500	42	31	41	57	85	131	116	115	88	110	46	47		
	PrecMin	222	1	1	0	0	4	5	3	6	11	6	1	0		
	PrecStd	72	10	8	10	15	19	31	29	27	19	23	12	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	2.7	9.1	24.2	30.8	33.4	33.8	33.0	28.9	19.8	8.9	2.5		
		MCWB	0.0	0.9	5.5	13.2	17.1	19.9	20.1	20.1	16.7	11.7	6.2	1.2		
	2%	DB	-1.3	0.4	4.9	20.3	28.0	31.2	31.2	30.1	25.0	16.4	6.2	0.8		
		MCWB	-2.0	-0.7	2.3	11.0	16.0	19.0	19.8	19.4	15.1	10.1	4.5	-0.1		
	5%	DB	-3.4	-1.8	3.1	17.2	25.4	29.1	29.2	27.5	21.7	13.7	4.2	-1.0		
		MCWB	-4.1	-2.7	1.1	9.7	14.9	18.8	19.5	18.6	14.3	8.8	2.8	-1.7		
	10%	DB	-5.8	-4.1	1.7	14.2	22.9	26.7	27.4	25.2	19.0	11.3	2.5	-2.9		
		MCWB	-6.4	-4.9	0.2	8.6	13.9	17.7	19.1	17.7	13.1	7.8	1.4	-3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	1.0	5.2	13.8	19.2	22.0	22.2	21.7	18.2	12.7	6.8	1.3		
		MCDB	0.6	2.1	8.4	21.6	28.7	30.3	28.5	28.6	26.5	17.7	8.1	2.0		
	2%	WB	-1.9	-0.6	2.6	11.9	17.0	20.7	21.3	20.6	16.1	10.9	4.6	0.0		
		MCDB	-1.3	0.4	4.4	18.7	24.6	28.1	28.2	27.7	22.5	15.3	5.9	0.9		
	5%	WB	-4.0	-2.7	1.5	10.4	15.8	19.5	20.6	19.4	14.8	9.3	3.0	-1.7		
		MCDB	-3.3	-1.9	2.8	16.6	23.8	26.6	27.4	25.5	20.6	12.8	4.1	-1.0		
	10%	WB	-6.4	-4.9	0.4	8.7	14.5	18.4	19.7	18.4	13.5	8.0	1.5	-3.6		
		MCDB	-5.8	-4.1	1.6	14.0	21.6	25.1	25.8	23.7	18.2	11.1	2.5	-3.0		
(35) Mean Daily Temperature Range	MDBR		7.3	8.7	8.7	10.1	11.9	11.3	10.9	10.5	9.8	7.6	6.3	6.7		
	5% DB	MCDBR	7.8	8.2	8.0	15.0	15.5	14.7	14.1	14.3	14.1	12.4	6.6	6.7		
		MCWBR	7.6	7.6	6.0	7.5	6.5	5.9	5.0	5.6	6.4	7.1	5.3	6.2		
	5% WB	MCDBR	7.9	8.1	7.5	13.6	14.1	13.4	12.2	12.4	12.6	11.1	6.6	6.5		
		MCWBR	7.8	7.6	5.9	7.3	6.6	5.8	5.2	5.5	6.4	7.0	5.6	6.1		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.320	0.376	0.416	0.416	0.410	0.429	0.443	0.390	0.360	0.320	0.289		
	taud		2.048	2.017	1.944	2.101	2.206	2.297	2.245	2.172	2.279	2.231	2.132	2.019		
	Ebn at Noon		591	717	761	797	825	836	809	765	758	675	550	483		
	Edh at Noon		67	101	139	138	132	122	126	128	100	82	62	55		
(44) All-Sky Solar Radiation	RadAvg		0.89	1.94	3.50	4.47	5.44	5.90	5.59	4.41	2.98	1.51	0.76	0.57		
	RadStd		0.06	0.09	0.20	0.48	0.47	0.52	0.46	0.43	0.25	0.25	0.10	0.07		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page